2D lid-driven cavity flow

Project Report 1

Group 19

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1 Statement of the Problem

The accuracy of a CFD simulation depends not only on the physics being modeled but also on the numerical choices made when setting up the solver. These choices are often made by habit or convention, and their individual effects, and especially their interactions, are rarely quantified together. The two-dimensional lid-driven cavity is a standard benchmark for studying this question because a widely used reference solution exists [1], which allows the error of any simulation to be measured directly.

The goal of this project is to use designed experiments to determine which solver settings have the greatest influence on prediction accuracy, whether the effect of one setting depends on another, and which combination of settings gives the lowest error.

2 Factors, Levels, and Ranges

The system is steady, incompressible, laminar flow in a unit square cavity whose top wall (the lid) moves at a constant speed. Four design factors are varied at two levels each, and Reynolds number is used as a blocking variable. Table 1 lists the factors and their levels. Lid speed, cavity size, and all other solver settings are held constant across runs, and the Reynolds number is set through the fluid viscosity.

Table 1: Design factors, levels, and block.

Factor	Description	Low (−)	High (+)
A	Mesh resolution	32×32	128×128
B	Convection scheme	First-order upwind	Second-order
C	Pressure–velocity coupling	SIMPLE	SIMPLEC
D	Convergence tolerance	10^{-3}	10^{-6}
Block	Reynolds number	100, 400, 1000	

The levels span the practical range of each setting, from a coarse, low-order, loosely converged configuration to a fine, higher-order, tightly converged one. Reynolds number is a blocking variable rather than a factor of interest: it changes the flow itself, and the goal is to find settings that perform well across these conditions rather than to study Reynolds number as a treatment.

The four factors at two levels each give $2^4 = 16$ treatment combinations, each of which is run once in every block, for a total of 48 runs.

3 Response Variable

The response is the prediction error of each simulation relative to the benchmark solution of Ghia et al. [1]. That reference tabulates the horizontal velocity u along the vertical centerline ($x = 0.5$) and the vertical velocity v along the horizontal centerline ($y = 0.5$) at a set of tabulated points for each Reynolds number.

For each run, the converged solution is sampled along both centerlines and interpolated to the benchmark point locations. The response E is the root-mean-square difference over all benchmark points, expressed as a percentage of the lid speed U_{lid}:

$$E = \frac{100}{U_{\text{lid}}} \sqrt{\frac{1}{N} \left[\sum_{i=1}^{N_u} \left(u_i^{\text{sim}} - u_i^{\text{ref}} \right)^2 + \sum_{j=1}^{N_v} \left(v_j^{\text{sim}} - v_j^{\text{ref}} \right)^2 \right]}, \tag{1}$$

where N_u and N_v are the numbers of benchmark points on each centerline and $N = N_u + N_v$. Lower values of E indicate a more accurate simulation, and each run produces one value of E.

References

[1] U. Ghia, K. N. Ghia, and C. T. Shin, "High-Re solutions for incompressible flow using the Navier–Stokes equations and a multigrid method," *Journal of Computational Physics*, vol. 48, no. 3, pp. 387–411, 1982.